

SWARNIM BARAPATRE

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with experience in developing mission-critical software for defense and enterprise organizations. Proven track record in designing systems and developing full-stack applications from scratch, regardless of the technology stack.

EDUCATION

BITS Pilani
Bachelor's in Chemical Engineering + Master's in Biological Sciences

Vasco da Gama, Goa
May, 2019 — July, 2024

PROFESSIONAL EXPERIENCE

Coditation Systems (Plant360.ai)
FullStack Software Engineer

Pune, Maharashtra
Feb, 2026 — Present

- Contributed to the Plant360.ai platform backend (Python/Django), developing and extending production API services.
- Reverse-engineered an undocumented third-party CAD SDK to programmatically extract complex drawing data into a canonical JSON data model, shipping the converter as both a CLI and a REST API.
- Designed and built bidirectional data-conversion pipelines around that canonical model, integrating multiple proprietary CAD formats with the DEXPI (ISO 15926) industry exchange standard and validating all generated output against the official Proteus 4.1.1 schema for correctness.
- Built a custom symbol-conversion tool that automated a previously manual drafting workflow across 90+ symbol definitions, cutting 2 hours/day of repetitive effort down to a verification step.

AccurateIC
FullStack Software Engineer, AI Department

Pune, Maharashtra
Aug, 2024 — Feb, 2026

- Developed full-stack application for AI-powered naval autonomous collision avoidance systems, handling real-time radar, camera & sensor data stream processing for 10+ vessels.
- Led full-stack development for **BlueGulf** x **AccurateIC** international collaboration, adapting autonomous navigation systems for high-speed vessels (80+ mph); successfully showcased at **DriftX Abu Dhabi Autonomous Expo** at Yas Marina.
- Architected, developed and led the development of building a platform for IOT-enabled systems (especially engines), smoothening the behavioural analysis of such systems, predicting failures and optimizing the systems for longevity using Artificial Intelligence.
- Drove technical decision-making and architecture planning towards making the software efficient, robust & maintainable.
- Identified performance bottlenecks & carried out code optimization.
- Mentored & trained junior developers and conducted code reviews of pull requests to maintain quality and correctness.
- Implemented standard development practices to bring in uniformity and long term maintainability of projects.

MasterSoft ERP Solutions
Software Engineering Intern, R&D Department

Nagpur, Maharashtra
Aug, 2023 — June, 2024

- Architected and developed products serving 700+ concurrent users as part of a core R&D team, ensuring optimal performance under high-load conditions.
- Developed Panopticon resulting in annual savings of Rs 53 Lakhs by eliminating dependency on third-party services and implementing in-house productivity measurement systems.
- Built comprehensive internal developer toolchains and automation frameworks, reducing development cycle times by up to 14 days and improving overall team productivity across multiple projects.
- Maintained critical Linux server infrastructure, implemented proactive monitoring solutions, and resolved production incidents to ensure system uptime and reliability.

SKILLS

- Languages:** TypeScript, JavaScript, Python, Go, SQL, Bash, Powershell, Pascal
- Frontend:** React, Tanstack Query, HTML, CSS, TailwindCSS

- **Backend:** AdonisJS MVC, ExpressJS, NodeJS, OAuth, RabbitMQ, Lucid ORM, Flask, FastAPI, Fiber, WebSockets, Server-Sent Events
- **Databases:** PostgreSQL, SQLite, Microsoft SQL Server, MongoDB
- **Cloud/DevOps:** AWS EC2, Nginx, Linux
- **Other:** Emacs, Vim, Git, LaTeX, Typst, GitLab API

PROJECTS

ISACA — Intelligent Situational Awareness & Collision Avoidance Ship Oct, 2024 — Present

- Architected and developed full-stack monorepo application (React, AdonisJS MVC, PostgreSQL) for autonomous vessel control, integrating ROS middleware via custom pub-sub data transport with real-time multi-sensor data fusion (Radar, GPS, Dual Cameras, eMRU, Odometry) for naval collision avoidance across 25+ ROS topics.
- Implemented radar overlay with obstacle trajectory prediction and MARPA tracking.
- Built route planning system with dynamic waypoint management, geodistance calculations, and voyage tracking across 12 feature modules (Radar, AIS, Battery, Engine, Routes, Sensors, etc.).
- Developed script execution system with real-time WebSocket monitoring and broadcasting.
- Implemented type-safe validation using VineJS across 11 RESTful controllers and 10 database migrations.
- Set up automated CI/CD pipeline with Jenkins for continuous deployment, including automated database migrations, seeding, and systemd service orchestration.
- Achieved 40% CPU reduction through performance profiling and code optimization.
- Designed bidirectional WebSocket architecture for shore-to-ship control synchronization.
- Contributed 97% of codebase (12K lines across 171 files).
- Tech Stack: *React, Redux, AdonisJS, Express JS, Lucid ORM, VineJS, PostgreSQL, Socket.io, ROS, WebRTC, Jenkins.*

NeuroGen — Smart Monitoring & Alert Systems for GenSets Dec, 2024 — Present

- Designed the system architecture and database structure.
- Features include: Predictive Maintenance, Multi-variate Anomaly Detection, Life Expectancy Simulations, Data Analysis & Visualizations.
- This application runs along with machine learning models in order to predict potential failures and provides alerts and recommendations.
- Led the development of the web application from scratch using a modern tech stack using *AdonisJS, Lucid ORM, ReactJS, Tanstack Query, and Tailwind CSS.*

Panopticon — Employee Performance Measurement System Oct, 2023 — June, 2024

- Designed and developed a multi-tenant application for streamlined employee productivity measurement.
- Reduced annual costs by Rs 53 Lakhs by eliminating reliance on external services (DeskTime).
- Facilitated confident remote talent recruitment through accurate productivity metrics.
- Implemented features such as interventionless atomic app updates, a live-updating configuration system, and encrypted activity logs.
- Tech Stack: *Python, Subprocess, Multi-threading, Pascal (Delphi), OpenCV, Powershell, Win32 APIs*

MS-SQL Suite of Tools Oct, 2023 — June, 2024

- Crafted a suite of tools for the Microsoft SQL Server meant for internal use by database developers, achieving 85% time reduction in SQL task flows.
- Developed a tool to optimize memory consumption by identifying memory leaks in the database, preventing database server crashes.
- Developed tools to compare and contrast database stored procedures and SQL functions, visualizing differences (side-by-side diff comprison) via automated reports and generating alter scripts to synchronize databases.
- Tech Stack: *Python, difflib, sqlparse, pyodbc*

Kafkaesque — Persistent Queue Management System Oct, 2023 — June, 2024

- Built a reliable queue system to streamline automation of time-intensive tasks, providing FIFO processing guarantees and remote task submission capabilities.
- Designed with persistence mechanisms to prevent data loss and configurable parallel processing modes, achieving significant performance improvements when order-independent processing is suitable.
- Addressed common challenges in task management workflows by providing a lightweight alternative to enterprise messaging systems for smaller-scale applications.
- Tech Stack: *Python, FastAPI, Pydantic*

OpenAI Question Paper & Course Outcome Generator Apr, 2023 — May, 2024

- Developed an AI-powered tool that automatically generates comprehensive course outcomes from course handouts and syllabi.
- Implemented question paper generation functionality with automatic classification based on Bloom's Taxonomy levels.
- Created RESTful API endpoints for seamless integration with existing educational management systems.
- Reduced manual effort in curriculum planning through automated course outcome generation.
- Tech Stack: *Python, OpenAI API, FastAPI, Pydantic*

EXTRA CURRICULARS

Swimming: National Swimmer. Multiple times State Gold Medalist.